LESSON 4: HOLOMORPHIC FUNCTIONAL CALCULUS

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In this lecture, we will investigate how to define functions of bounded linear operators. We start with the simplest case, the polynomial functional calculus. We then mention the power series expansions of holomorphic functions to define the functions of a bounded linear operator. Finally, we use the Cauchy formula to provide definitions of holomorphic functional calculus and study its main properties.

1. Elementary Functional Calculi

Definition 1.1 (The Polynomial Functional Calculus)**.** *Let X be a Banach space and $A : X \to X$ be a bounded linear operator.*

- *The polynomial functional calculus allows us to define a polynomial in A for any polynomial p.*
- *Consider a polynomial p of the form:*
$$p(\lambda) = a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$$
where $a_0, a_1, \ldots, a_n$ are scalars in $\mathbb{C}$ with $a_n \neq 0$.
- *The polynomial functional calculus is given by:*
$$p(A) = a_nA^n + a_{n-1}A^{n-1} + \cdots + a_1A + a_0I$$
where I is the identity operator on X.

Theorem 1.2. *Properties For any polynomial p and q, and scalars α and β in $\mathbb{C}$ we have:*

- *Linearity:*
$$(\alpha p + \beta q)(A) = \alpha p(A) + \beta q(A).$$
- *Multiplicativity: For any polynomials p and q:*
$$(pq)(A) = p(A)q(A)$$
- *Identity: The constant polynomial 1 gives:*
$$1(A) = I$$

Example 1.3. *Example Let $p(\lambda) = \lambda^2 + 2\lambda + 3$. Then: $p(A) = A^2 + 2A + 3I$.*

Remark 1.4. *The proof is direct and it is a useful exercise.*

Definition 1.5 (Functional calculus for powers series expansion)**.** *Let X be a Banach space and $A : X \to X$ be a bounded linear operator.*

- *Suppose f is a function that can be represented by a power series with radius of convergence $R = +\infty$, centered at 0:*
$$f(z) = \sum_{n=0}^{\infty} c_n z^n,$$
where c_n are complex numbers.

- *The functional calculus for f with respect to the bounded linear operator A is defined by:*
$$f(A) = \sum_{n=0}^{\infty} c_n A^n,$$
where the series converges in the operator norm.

Remark 1.6. *An example of application of the functional calculus for powers series expansion can be applied for the exponential function $f(z) = e^z$, which has the power series expansion:*
$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$
The functional calculus applied to $A \in \mathcal{B}(X)$, where X is a Banach space, is:
$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}$$
where the convergence is in the operator norm of $\mathcal{B}(X)$.

Remark 1.7. *For more information on the matrix exponential and its applications to linear systems, see Appendix A.*

Remark 1.8 (Time Evolution and the Exponential Map)**.** *The time evolution of a quantum state $\psi(t)$ under a time-independent Hamiltonian H is given by the Schrödinger equation:*
$$i\hbar \frac{d}{dt}\psi(t) = H\psi(t).$$
The solution can be expressed using the exponential of the Hamiltonian operator:
$$\psi(t) = e^{-iHt/\hbar}\psi(0).$$

The operator $e^{-iHt/\hbar}$ is defined via a suitable functional calculus. Here we cannot use the power series expansion because the operator $H : D(H) \subset X \to X$ is in general unbounded.

Instead of the cauchy kernel, use the resolvent operator

2. Properties of the spectrum of a bounded linear operator

Remark 2.1. *Since the resolvent operator is a holomorphic function that is operator-valued, we summarize the main facts regarding holomorphic operators in Appendix B.*

Remark 2.2. *Here we recall the resolvent equation, but for the purpose of the holomorphic functional calculus the resolvent operators is given by*
$$R(\lambda, A) := (\lambda\mathcal{I} - A)^{-1}$$
instead of $(A - \lambda\mathcal{I})^{-1}$ because it better fit with the Cauchy formula. Nothing change on the property of the spectrum.

Remark 2.3. *Often, when no confusion arises, the operator $(\lambda\mathcal{I} - A)^{-1}$ is simply denoted as $(\lambda - A)^{-1}$, where λ is implicitly understood to represent $\lambda\mathcal{I}$.*

Definition 2.4. *Let X be a complex Banach space and $A \in \mathcal{B}(X)$. We give the following definitions.*

(1) *The resolvent set $\rho(A)$ of A is the set of complex numbers λ for which $(\lambda\mathcal{I} - A)^{-1}$ exists as a linear bounded operator with domain X.*
(2) *The spectrum $\sigma(A)$ of A is the complement of $\rho(A)$.*
(3) *The function*
$$R(\lambda, A) = (\lambda - A)^{-1},$$
defined on $\rho(A)$, is called the resolvent operator of A.
(4) *The number*
$$r(A) = \sup\{|\lambda| : \lambda \in \sigma(A)\}$$
is called the spectral radiusof A.

Remark 2.5. *The intuition on which the holomorphic functional calculus is based is replacing the Cauchy kernel in the Cauchy formula with the resolvent operators and integrating around the spectrum of A. The motivation is that*
$$(\lambda\mathcal{I} - A)^{-1} : \rho(A) \to \mathcal{B}(V)$$
is a holomorphic function.

In doing so, we need to pay attention to the following facts:

(1) *The Cauchy formula works because the point z_0 is inside the closed curve ∂V.*
(2) *In the case where the point would be outside the curve ∂V, the integral would be zero (Cauchy theorem).*

So, replacing $(\lambda - z_0)^{-1}$ by $(\lambda\mathcal{I} - A)^{-1}$, we have to be sure that:

- *The spectrum is compact (closed and bounded in $\mathbb{C}$).*
- *The spectrum of a bounded linear operator is nonempty.*

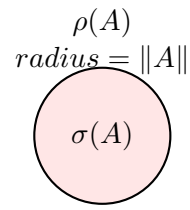
The following theorem assures the above requirements.

Theorem 2.6 (Spectrum of a bounded operator)**.** *Let $A \in \mathcal{B}(X)$, where $X \neq \{0\}$.*

(1) *The set $\sigma(A)$ is compact (Closed and bounded).*
(2) *The set $\sigma(A)$ is non-empty.*

Proof. Spectrum of a bounded operator:

(1A) We proved that: The set $\sigma(A)$ is closed because the resolvent set $\rho(A)$ is open.
(1B) The set $\sigma(A)$ is bounded since for $\lambda \in \mathbb{C}$ such that $|\lambda| > \|A\|$ we have
$$(\lambda - A)^{-1} = \lambda^{-1}(\mathcal{I} - \frac{1}{\lambda}A)^{-1} = \sum_{n=0}^{\infty} \lambda^{-n-1}A^n$$
since $|\lambda|^{-1}\|A\| < 1$ so the spectrum $\sigma(T)$ is inside the region of the complex plane $|\lambda| \leq \|A\|$ and so it is bounded.



(2) Closed and bounded implies compact in $\mathbb{C}$.

The set $\sigma(A)$ is non-empty. Suppose by absurd that $\sigma(A) = \emptyset$.

Then the function

$$(\lambda - A)^{-1} : \rho(A) \to \mathcal{B}(X)$$

is analytic (holomorphic) in all complex plane but from the inequality (in which we use the property $\|A^n\| \leq \|A\|^n$)

$$\begin{aligned}
\|(\lambda - A)^{-1}\| &= \left\| \sum_{n=0}^{\infty} \lambda^{-n-1} A^n \right\| \\
&\leq \sum_{n=0}^{\infty} \|\lambda^{-n-1} A^n\| \\
&\leq \sum_{n=0}^{\infty} |\lambda|^{-n-1} \|A\|^n \\
&= |\lambda|^{-1} \sum_{n=0}^{\infty} \left(|\lambda|^{-1} \|A\|\right)^n \\
&= \frac{1}{|\lambda|} \cdot \frac{1}{1 - |\lambda|^{-1} \|A\|} \\
&= \frac{1}{|\lambda| - \|A\|}, \quad \text{for} \quad |\lambda| > \|A\|,
\end{aligned}$$

we have

$$\lim_{\lambda \to \infty} \|(\lambda - A)^{-1}\| = 0$$

since a function that is entire in $\mathbb{C}$ it has to be constant (Liouville's theorem) and since $(\lambda - A)^{-1}$ at infinity is zero it has to be $(\lambda - A)^{-1} = 0$, but this cannot be since $(\lambda - A)^{-1}$ is the inverse of $\lambda - A$. So the spectrum cannot be nonempty.

□

Remark 2.7 (The resolvent operator we will use for the holomorphic functional calculus)**.** *Let $R(\lambda, A) := (\lambda \mathcal{I} - A)^{-1}$. For every pair λ, $\mu \in \rho(A)$ we have*

(1)

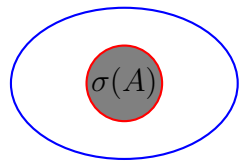
$$R(\lambda, A) R(\mu, A) = R(\mu, A) R(\lambda, A).$$

(2) *(Resolvent equation)*

$$R(\lambda, A) - R(\mu, A) = (\mu - \lambda) R(\lambda, A) R(\mu, A).$$

3. Definition of the Holomorphic Functional Calculus and Its Properties

Definition 3.1 (Definition of Analytic Functions on the spectrum)**.** *Let $A \in \mathcal{B}(X)$. By $\mathcal{F}(A)$ we denote the family of functions f which are analytic on some neighborhood of $\sigma(A)$.*



Definition 3.2. *Definition of the Riesz-Dunford functional calculus*

- *Let $f \in \mathcal{F}(A)$,*
- *let U be an open set whose boundary ∂U is a finite union of continuously differentiable Jordan curves, oriented in the positive sense.*
- *Suppose that $\sigma(A) \subset U$ and that $U \cup \partial U$ is contained in the domain of analyticity of f.*

Then we define the operator $f(A)$ as

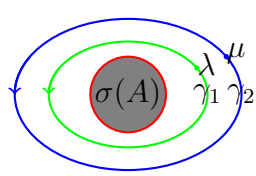
$$f(A) := \frac{1}{2\pi i} \int_{\partial U} R(\lambda, A)\, f(\lambda)\, d\lambda. \tag{3.1}$$

The holomorphic functional calculus is well-defined because the integral is independent of the contour of integration, as stated by the next theorem. The proof is fundamentally based on the fact that the resolvent operator is a holomorphic function that is operator-valued, along with the Cauchy formula.

Theorem 3.3. *Under the hypothesis in the definition of the Riesz-Dunford functional calculus the integral*

$$\frac{1}{2\pi i} \int_{\partial U} R(\lambda, A)\, f(\lambda)\, d\lambda$$

depends only on f and does not depend on the open set U.



Remark 3.4.
(I) ∂U can be for example γ_1 or γ_2 or any other curve that does not intersect $\sigma(A)$
(II) $\sigma(A) \subset U$
(III) $U \cup \partial U$ is contained in the domain of analyticity W of $f : W \subseteq \mathbb{C} \mapsto \mathbb{C}$

3.1. Properties of the holomorphic functional calculus.

Theorem 3.5 (Algebraic Properties of the functional calculus)**.** *Let f, $g \in \mathcal{F}(A)$, α_1, and let $\alpha_2 \in \mathbb{C}$. Then*

(1) *(Linearity)* $\alpha_1 f + \alpha_2 g \in \mathcal{F}(A)$ *and* $(\alpha_1 f + \alpha_2 g)(A) = \alpha_1 f(A) + \alpha_2 g(A)$.
(2) *(Product rule)*

$$f \cdot g \in \mathcal{F}(A) \quad \textit{and} \quad f(A)g(A) = (f \cdot g)(A).$$

(3) *(Compatibility with series expansions) If*

$$f(\lambda) = \sum_{n \geq 0} \alpha_n \lambda^n$$

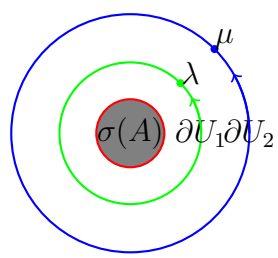
converging in a neighborhood of $\sigma(A)$*, then*

$$f(A) = \sum_{n \geq 0} \alpha_n A^n.$$

Proof. **Point (1)** It follows from the definition of the functional calculus.

Point (2) Since $f, g \in \mathcal{F}(A)$, it is obvious that $f \cdot g \in \mathcal{F}(A)$. Let U_1 and U_2 be two neighborhoods of $\sigma(A)$ whose boundaries ∂U_1 and ∂U_2 are finite unions of continuously differentiable Jordan curves. Let us assume that:

(a) $U_1 \cup \partial U_1 \subseteq U_2$.
(b) $U_2 \cup \partial U_2$ is contained in a common region of analyticity of f and g.



We will use the definition and the functional calculus for $f(A)$ and for $g(A)$ and the resolvent equation

$$R(\lambda, A)R(\mu, A) = \frac{R(\lambda, A) - R(\mu, A)}{\mu - \lambda}$$

where we observe that $\frac{1}{\mu-\lambda}$ is the Cauchy kernel. So $f(A)g(A)$ is given by

$$\begin{aligned}
f(A)g(A) &= -\frac{1}{4\pi^2}\int_{\partial U_1} f(\lambda)R(\lambda,A)d\lambda \int_{\partial U_2} g(\mu)R(\mu,A)d\mu \\
&= -\frac{1}{4\pi^2}\int_{\partial U_1}\int_{\partial U_2} f(\lambda)g(\mu)R(\lambda,A)R(\mu,A)d\mu d\lambda \\
&= -\frac{1}{4\pi^2}\int_{\partial U_1}\int_{\partial U_2} f(\lambda)g(\mu)\frac{R(\lambda,A)-R(\mu,A)}{\mu-\lambda}d\mu d\lambda \\
&= -\frac{1}{4\pi^2}\int_{\partial U_1} f(\lambda)R(\lambda,A)\left(\int_{\partial U_2}\frac{g(\mu)}{\mu-\lambda}d\mu\right)^{\clubsuit} d\lambda \\
&\quad +\frac{1}{4\pi^2}\int_{\partial U_2} g(\mu)R(\mu,A)\left(\int_{\partial U_1}\frac{f(\lambda)}{\mu-\lambda}d\lambda\right)^{\spadesuit} d\mu \\
&= \frac{1}{2\pi i}\int_{\partial U_1} f(\lambda)\Big(g(\lambda)\Big)^{\clubsuit} R(\lambda,A)d\lambda \\
&= (fg)(A),
\end{aligned}$$

where by the Cauchy formula we have

$$\left(\int_{\partial U_2}\frac{g(\mu)}{\mu-\lambda}d\mu\right)^{\clubsuit} = g(\lambda)$$

and where, by the Cauchy theorem we have

$$\left(\int_{\partial U_1}\frac{f(\lambda)}{\mu-\lambda}d\lambda\right)^{\spadesuit} = 0$$

because of the Cauchy theorem since μ belong to ∂U_2 so $\frac{f(\lambda)}{\mu-\lambda}$ is holomorphic. The lat term

$$\frac{1}{2\pi i}\int_{\partial U_1} f(\lambda)\Big(g(\lambda)\Big)^{\clubsuit} R(\lambda,A)d\lambda$$

is exactly $(fg)(A)$. So be have proved the product rule $f(A)g(A) = (fg)(A)$.

Point (3) Point (3) follows from the fact that the series $\sum_{n=0}^{\infty}\alpha_n\lambda^n$ converges uniformly on the set

$$C_\varepsilon = \{\lambda \in \mathbb{C} : |\lambda| = r(A)+\varepsilon\}$$

for suitable $\varepsilon > 0$. We have, for the uniform convergence

$$\begin{aligned} f(A) &= \frac{1}{2\pi i}\int_{C_\varepsilon}\sum_{n=0}^{\infty}\alpha_n\lambda^n R(\lambda, A)d\lambda = \frac{1}{2\pi i}\sum_{n=0}^{\infty}\alpha_n\int_{C_\varepsilon}\lambda^n R(\lambda, A)d\lambda \\ &= \frac{1}{2\pi i}\sum_{n=0}^{\infty}\alpha_n\int_{C_\varepsilon}\left(\sum_{k=0}^{\infty}\lambda^{-1-k}A^k\right)\lambda^n d\lambda \\ &= \sum_{n=0}^{\infty}\sum_{k=0}^{\infty}\alpha_n\Big(\frac{1}{2\pi i}\int_{C_\varepsilon}\lambda^{-1-k+n}d\lambda\Big)A^k \\ &= \sum_{n=0}^{\infty}\alpha_n A^n. \end{aligned}$$

We compute the integral

$$\frac{1}{2\pi i}\int_{C_\varepsilon}\lambda^{-1-k+n}d\lambda \tag{3.2}$$

Set

$$R := r(A) + \varepsilon$$

in polar coordinates $\lambda = Re^{i\theta}$, so that $d\lambda = Rie^{i\theta}d\theta$. Then, we have:

$$\frac{1}{2\pi i}\int_{C_\varepsilon}\lambda^{-1-k+n}d\lambda = \frac{1}{2\pi i}\int_0^{2\pi}(Re^{i\theta})^{-1-k+n}(Rie^{i\theta})d\theta.$$

$$= \frac{1}{2\pi i}R^{-1-k+n}Ri\int_0^{2\pi}e^{i(-k+n)\theta}d\theta.$$

Now, we can evaluate the integral:

$$\int_0^{2\pi}e^{i(-k+n)\theta}d\theta.$$

This integral evaluates to:

$$\begin{cases} 2\pi & \text{if } -k+n=0 \\ 0 & \text{otherwise} \end{cases}$$

Thus, summing up the results:

$$\frac{1}{2\pi i}R^{-k+n}i\cdot\begin{cases} 2\pi & \text{if } n=k \\ 0 & \text{otherwise.} \end{cases}$$

□

Theorem 3.6 (Spectral mapping theorem for A^n)**.** *Let $A \in \mathcal{B}(X)$, where $X \neq \{0\}$. Then we have*

$$\sigma(A^n) = [\sigma(A)]^n := \{\lambda^n : \lambda \in \sigma(A)\}.$$

Proof. We proof two inclusions that give the statement. We have to show that μ belongs to the spectrum of A^n if and only if

$$\mu = \lambda^n, \quad \lambda \in \sigma(A).$$

- We first prove $\sigma(A^n) \supseteq [\sigma(A)]^n$.

For n integer recall the factorization

$$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \ldots + ab^{n-2} + b^{n-1}).$$

Since A and its powers commute we can write

$$\lambda^n \mathcal{I} - A^n = (\lambda\mathcal{I} - A)(\lambda^{n-1}\mathcal{I} + \lambda^{n-2}A + \ldots + A^{n-1}).$$

This shows that if $\lambda\mathcal{I} - A$ is not invertible, then $\lambda^n\mathcal{I} - A^n$ is also not invertible. This proves $\sigma(A^n) \supseteq [\sigma(A)]^n$.

- Now we prove $\sigma(A^n) \subseteq [\sigma(A)]^n$.

Suppose the $\nu \notin [\sigma(A)]^n$. By the Fundamental Theorem of Algebra we have

$$(\nu - \lambda^n) = (-1)^{n+1}(\lambda_1 - \lambda)(\lambda_2 - \lambda)\ldots(\lambda_n - \lambda),$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the roots of ν and $\lambda_i \neq \lambda_j$ if and only if $i \neq j$.

By applying the product rule:

$$(fg)(A) = f(A)g(A),$$

we write:

$$(\nu\mathcal{I} - A^n) = (-1)^{n+1}(\lambda_1\mathcal{I} - A)(\lambda_2\mathcal{I} - A)\ldots(\lambda_n\mathcal{I} - A).$$

Since $\nu\mathcal{I} - A^n$ is not invertible, there exists an i such that $\lambda_i\mathcal{I} - A$ is not invertible. This implies $\lambda_i \in \sigma(A)$, hence $\sigma(A^n) \subseteq [\sigma(A)]^n$. □

Theorem 3.7 (Spectral radius theorem (Gelfand))**.** *Let $A \in \mathcal{B}(X)$. Then we have*

$$r(A) = \lim_{n\to\infty} \sqrt[n]{\|A^n\|}$$

where the spectral radius is defined as

$$r(A) = \sup\{|\lambda| : \lambda \in \sigma(A)\}.$$

Proof. We will show that

$$r(A) \leq \liminf_{k\to\infty} \|A^k\|^{1/k}$$

and

$$\limsup_{n\to\infty} \|A^n\|^{1/n} \leq r(A)$$

which also imply the existence of the limit. Recall that $\rho(A)$ is open so $\sigma(A)$ is closed.

We reconsider the resolvent operator series. Precisely, let k be an integer so we can always decompose $n = kq + r$ for suitable $q \in \mathbb{N}$ and $0 \leq r < k$. So

$$A^n = A^{kq+r} = (A^k)^q A^r$$

so

$$\|A^n\| \leq \|A^r\| \|(A^k)\|^q.$$

The resolvent series is given by

$$(\lambda - A)^{-1} = \sum_{n=0}^{\infty} \lambda^{-1-n} A^n$$

and

$$\|\sum_{n=0}^{\infty} \lambda^{-1-n} A^n\| \leq \sum_{n=0}^{\infty} |\lambda|^{-1-n} \|A^n\|$$
$$\leq \Big(\sum_{r=0}^{k-1} |\lambda|^{-1-r} \|A^r\| \Big) \Big(\sum_{q=0}^{\infty} (|\lambda|^{-k} \|(A^k)\|)^q \Big).$$

The series converges absolutely if

$$1 > \|A^k\|/|\lambda|^k \quad \Rightarrow \quad |\lambda| > \|A^k\|^{1/k}.$$

It follows that every $\mu \in \sigma(A)$ must satisfy

$$|\mu| \leq \|A^k\|^{1/k}$$

since this holds for all integer k

$$r(A) := \lim_{k\to\infty} \sqrt[k]{\|A^k\|} \leq \liminf_{k\to\infty} \|A^k\|^{1/k}.$$

Now, using the Cauchy formula and the functional calculus for A^n we have

$$A^n = \frac{1}{2\pi i} \int_{\partial U} (\lambda - A)^{-1} \lambda^n \, d\lambda$$

where for the path of integration ∂U we take the circle $|\lambda| = r(A) + \delta$ for $\delta > 0$. We can estimate the operator as:

$$\|A^n\| \leq c(r(A) + \delta)^{n+1}, \quad c := \max_{|\lambda| = r(A)+\delta} \|(\lambda - A)^{-1}\|$$

Take the n-th root:

$$\|A^n\|^{1/n} \leq c^{1/n} (r(A) + \delta)^{(n+1)/n}$$

and then we take the supremum of

$$\|A^n\|^{1/n} \leq c^{1/n} (r(A) + \delta)^{(n+1)/n}$$

so that we get

$$\limsup_{n\to\infty} \|A^n\|^{1/n} \leq (r(A) + \delta)$$

Since this is true for every δ, we take the limit for $\delta \to 0$:

$$\limsup_{n\to\infty} \|A^n\|^{1/n} \leq r(A).$$

This concludes the proof. □

Example 3.8. *Example of spectral mapping theorem*

- $\sigma_p(A) = \{\lambda_1 = 2,\ \lambda_2 = 3\}$ *is the spectrum of the operator* $A : \mathbb{R}^2 \to \mathbb{R}^2$
$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$
- *For* $P(z) = z^2 - 3z + 1$, *we have that* $P(A) = A^2 - 3A + I$ *is given by*
$$P(A) = \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix}$$
- *The spectrum of the operator* $P(A) : \mathbb{R}^2 \to \mathbb{R}^2$ *is*
$$\sigma_p(P(A)) = \{\mu_1 = -1,\ \mu_2 = 1\}$$
- *Observe that* $P(\lambda_1) = P(2) = -1 = \mu_1$, *and* $P(\lambda_2) = P(3) = 1 = \mu_2$ *That is,*
$$\sigma(\lambda_j) = \mu_j \quad \textit{for} \quad j = 1, 2 \quad \Rightarrow P(\sigma(A)) = \sigma(P(A)).$$

Theorem 3.9 (General version of the Spectral Mapping Theorem)**.** *If* $f \in \mathcal{F}(A)$, *then*
$$f(\sigma(A)) = \sigma[f(A)].$$

Proof. We remark that we have to show that μ belongs to the spectrum of $f(A)$ if and only if
$$\mu = f(\lambda), \quad \lambda \in \sigma(A).$$
- If μ is not of the form $\mu = f(\lambda)$ for $\lambda \in \sigma(A)$ then
$$\xi \mapsto f(\xi) - \mu$$
does not vanish on $\sigma(A)$.
- Therefor the function $g(\xi) := (f(\xi) - \mu)^{-1}$ is analytic in an open set that contains $\sigma(A)$. Using the functional calculus we define $g(A)$ so
$$g(A) := (f(A) - \mu)^{-1}$$
But also using the product rule of the functional calculus for $(f(\xi) - \mu)g(\xi) := 1$ we have
$$(f(A) - \mu I)g(A) = I$$
so $g(A)$ is the inverse of $(f(A) - \mu I)$ this implies that $\mu \notin \sigma(f(A)$.

Conversely, suppose that μ is of the form $\mu = f(\lambda)$ and define the function
$$\kappa(\xi) = \frac{f(\xi) - f(\lambda)}{\xi - \lambda},$$
where κ is analytic in an open set that contains $\sigma(A)$ so we define $\kappa(A)$ via the functional calculus.

- So writing $(\xi - \lambda)\kappa(\xi) = f(\xi) - f(\lambda)$ and using the product rule of the functional calculus, we also have
$$(A - \lambda I)\kappa(A) = f(A) - f(\lambda)I$$

- Now if $f(A) - f(\lambda)\mathcal{I}$ had a bounded everywhere defined inverse $[f(A) - f(\lambda)\mathcal{I}]^{-1}$, then $(A - \lambda I)\kappa(A)$ would be a bounded everywhere defined inverse of $(A - \lambda I)$.

This implies that $f(\lambda) \in \sigma[f(A)]$. □

Theorem 3.10 (Composition of functions for the functional calculus)**.** *Let $f \in \mathcal{F}(A)$, $g \in \mathcal{F}(f(A))$, and $F(\lambda) = g(f(\lambda))$. Then we have:*

(1) $F \in \mathcal{F}(A)$,
(2) $F(A) = g(f(A))$.

Remark. It is a consequence of the spectral mapping theorem. We omit the proof.

Theorem 3.11 (Perturbation of the functional calculus)**.** *Let A be a linear bounded operator, $f \in \mathcal{F}(A)$, and $\varepsilon > 0$. Then there exists a $\delta > 0$ such that*

- *if A_1 is a bounded operator and*
- $\|A_1 - A\| < \delta$,

then $f \in \mathcal{F}(A_1)$ and $\|f(A_1) - f(A)\| < \varepsilon$.

Remark. We omit the proof.

4. Appendix A: The Matrix Exponential

Definition 4.1 (Matrix Exponential)**.** *The matrix exponential is a fundamental concept in linear algebra and differential equations.*

- *For a square matrix $A \in \mathbb{R}^{n\times n}$, the matrix exponential e^A is defined by the following series:*
$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!},$$
where A^0 is the identity matrix I, and A^k denotes the k-th power of the matrix A.
- *The convergence is in the norm of the spaces of matrices*
$$\|A\| = \sum_{i,j=1}^{n} |a_{i,j}|, \quad \text{where the elements of } A \text{ are} \quad \{a_{i,j}\}_{i,j=1}^{n}.$$

Theorem 4.2. *Properties of the matrix exponential:*

(1) *Identity Property: The exponential of the zero matrix is the identity matrix:*
$$e^0 = I.$$
(2) *Product of Exponentials: If A and B commute, i.e., $AB = BA$, then:*
$$e^{A+B} = e^A e^B.$$
(3) *Derivative: The derivative of the matrix exponential with respect to t is given by:*
$$\frac{d}{dt} e^{tA} = Ae^{tA}.$$
(4) *Inverse: The inverse of the matrix exponential is the exponential of the negative matrix:*
$$(e^A)^{-1} = e^{-A}.$$

Theorem 4.3 (Linear Systems of Ordinary Differential Equations)**.** *Consider a linear system of ordinary differential equations (ODEs) given by:*

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x},$$

where $\mathbf{x}(t)$ *is an* n*-dimensional vector and* A *is a constant* $n \times n$ *matrix. Then, the solution to this system can be expressed in terms of the matrix exponential as follows:*

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0),$$

where $\mathbf{x}(0)$ *is the initial condition vector.*

Example 4.4. *Consider the system:*

$$\frac{d}{dt}\mathbf{x} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\mathbf{x} \quad \text{with} \quad \mathbf{x}(0) = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

The matrix A *is:*

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

To find e^{tA}*, (with some computations):*

$$e^{tA} = e^{t\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}} = \begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}.$$

Thus, the solution to the system is:

$$\mathbf{x}(t) = \begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}.$$

5. Appendix B: Vector Valued Functions of a Complex Variable

The theory of holomorphic functions plays an important role in mathematics, particularly in operator theory, where it is crucial for the definition of functions of operators. Here we recall just some basic facts of this theory for functions from the complex numbers to a Banach space.

Definition 5.1 (Holomorphic Functions with values in a Banach space)**.** *Let* X *be a Banach space,* $f : U \subseteq \mathbb{C} \to X$ *be a function and let* $z_0 \in U$*. We say that* f *is holomorphic in* z_0 *if there exists an open disc* $D(z_0, r)$*,* $r > 0$ *contained in* U*, such that* f *admits the power series expansion*

$$f(z) = \sum_{n \geq 0} A_n (z - z_0)^n, \quad A_n \in X, \quad n \in \mathbb{N},$$

converging in the norm of X *in* $D(z_0, r)$*.*

Remark 5.2 (Cauchy Theorem and Hahn-Banach theorem)**.** *The classical Cauchy theorems can be generalized to functions with values in a normed vector space X. To this purpose, we will make use of the following result which is a corollary of the Hahn-Banach theorem:*

Corollary 5.3 (Corollary of Hahn-Banach Theorem)**.** *Let X be a Banach space and let $x \in X$. If for any continuous linear functional $x' \in X'$ acting on X it is $\langle x, x' \rangle = 0$, then $x = 0$.*

This corollary says that the linear functionals x' in X' separates the points x on the Banach space X.

Theorem 5.4 (Cauchy Theorem)**.** *Let U be an open bounded set in $\mathbb{C}$ such that ∂U is a finite union of continuously differentiable Jordan curves.*

- *Let X be a Banach space.*
- *Let $f : W \subset \mathbb{C} \to X$ be a holomorphic function such that $U \cup \partial U \subset W$.*

Then

$$\int_{\partial U} f(z)dz = 0.$$

Remark 5.5. *The proof of the Cauchy Theorem is based on the fact that for every bounded linear functional x' on X the duality $\langle f, x' \rangle$ is holomorphic on $U \cup \partial U$ so by the Cauchy theorem for complex valued functions*

$$\langle \int_{\partial U} f d\eta, x' \rangle = \int_{\partial U} \langle f, x' \rangle d\eta = 0.$$

By the corollary of the Hahn-Banach theorem we get

$$\int_{\partial U} f d\eta = 0.$$

Theorem 5.6 (Cauchy Integral Formula)**.** *Let W be an open bounded set in $\mathbb{C}$.*

- *Let X be a Banach space.*
- *Let $f : W \to X$ be a holomorphic function.*
- *Suppose that $V \subset W$ such that $\partial V \cup V \subseteq W$ and such that ∂V is a finite union of continuously differentiable Jordan curves.*

Then, for each $z_0 \in V$ we have

$$f(z_0) = \frac{1}{2\pi i} \int_{\partial V} f(z)(z - z_0)^{-1} dz.$$

Theorem 5.7 (Liouville's theorem)**.** *Every holomorphic function $f : \mathbb{C} \to X$ such that there exists a positive number M such that*

$$\|f(z)\| \leq M, \qquad \forall z \in \mathbb{C}$$

must be is constant.